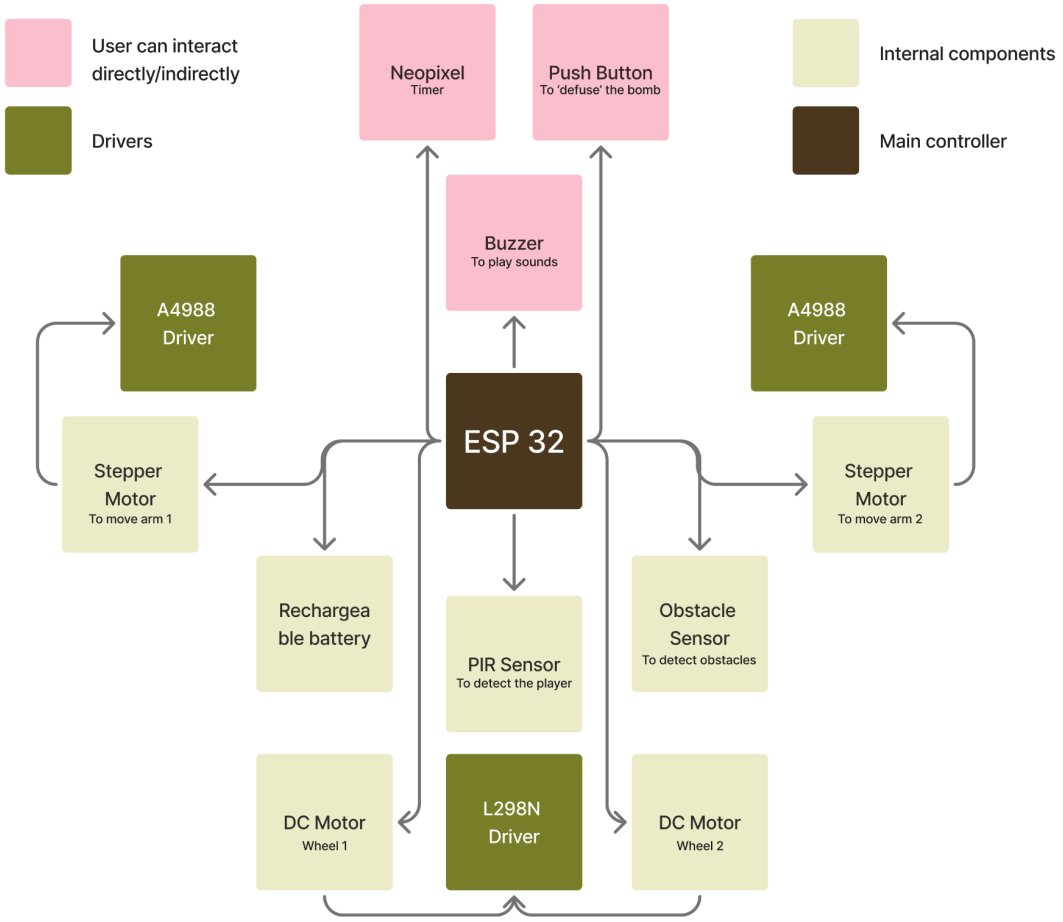
Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
ESP 32	*1*	*Central brain of the robot*	*Has built-in Bluetooth, dual-core CPU, and sufficient GPIO pins*
Stepper Motor	*2*	*To move the arms*	*Precise angle is required for this application and hence stepper motor*
DC Motor + Driver	*2*	*To drive the wheels*	*Continuous rotation with PWM speed control suits wheel movement.*
Neopixel	*1*	*To display game timer via colour*	*Fully RGB and addressable via a single pin.*
Buzzer	*1*	*To produce game sounds*	*Passive buzzer plays varied tones using PWM on one GPIO pin.*
PIR Sensor	*1*	*To detect player proximity (~3 m)*	*Detects body heat and changes the behaviour of SCURR-E accordingly*
IR Sensor	*1*	*To detect obstacles while moving*	*Provides short-range detection to prevent collisions.*
Push Button	*1*	*To act as the defuse button*	*Simple contact input is sufficient for the tag mechanic.*